

Name: _____ Register No: _____

Experiment 13

Question: Experiment to handle missing values and inappropriate data in a given data set.

Aim

To handle missing values and inappropriate data in a given data set.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def exp_13():
    df = pd.DataFrame({
        "Name": ["Asha", "Ravi", "Meena", "Kumar", "Divya", "Arun", "Priya", "Vijay"],
        "Age": [25, np.nan, 31, -5, 28, 250, 35, 29],
        "Salary": [50000, 42000, "N/A", 38000, "?", 70000, 55000, np.nan],
        "City": ["chennai", "CHENNAI ", "Mumbai", np.nan, "Delhi", "mumbai", "Delhi", "Chennai"],
        "Gender": ["F", "male", "Female", "M", "f", "M", np.nan, "Male"],
    })
    print("Original data:\n", df, "\n")
    print("Salary dtype before cleaning:", df["Salary"].dtype)

    # Step 1: convert inappropriate text in numeric column to NaN
    df["Salary"] = pd.to_numeric(df["Salary"], errors="coerce")
    # Step 2: inappropriate values (age must be 0-100) -> NaN
    df.loc[(df["Age"] < 0) | (df["Age"] > 100), "Age"] = np.nan
    print("Missing values after marking invalid data:\n", df.isnull().sum(), "\n")
    miss_before = df.isnull().copy()

    # Step 3: standardise text
    df["City"] = df["City"].str.strip().str.title()
    df["Gender"] = df["Gender"].str.strip().str.lower().str[0].map({"m": "Male", "f": "Female"})

    # Step 4: handle missing values
    df["Age"] = df["Age"].fillna(df["Age"].median()) # numeric -> median
    df["Salary"] = df["Salary"].fillna(df["Salary"].mean()) # numeric -> mean
    df["City"] = df["City"].fillna(df["City"].mode()[0]) # categorical -> mode
    df["Gender"] = df["Gender"].fillna(df["Gender"].mode()[0])

    print("Cleaned data:\n", df.round(1), "\n")
    print("Missing values after cleaning:\n", df.isnull().sum().to_dict())
    rows_left = int((~miss_before.any(axis=1)).sum())
    print("Alternative: dropping every row with a missing value would leave only",
          rows_left, "of", len(df), "rows")

    plt.figure(figsize=(6, 4))
    sns.heatmap(miss_before, cbar=False, cmap="viridis", yticklabels=False)
    plt.title("Missing values before cleaning (yellow = missing)")
    plt.show()
```

exp_13()

Output

Original data:

	Name	Age	Salary	City	Gender
0	Asha	25.0	50000	chennai	F
1	Ravi	NaN	42000	CHENNAI	male
2	Meena	31.0	N/A	Mumbai	Female
3	Kumar	-5.0	38000	NaN	M
4	Divya	28.0	?	Delhi	f
5	Arun	250.0	70000	mumbai	M
6	Priya	35.0	55000	Delhi	NaN
7	Vijay	29.0	NaN	Chennai	Male

Salary dtype before cleaning: object
Missing values after marking invalid data:

Name	0
Age	3
Salary	3

```
City      1
Gender    1
dtype: int64
```

Cleaned data:

	Name	Age	Salary	City	Gender
0	Asha	25.0	50000.0	Chennai	Female
1	Ravi	29.0	42000.0	Chennai	Male
2	Meena	31.0	51000.0	Mumbai	Female
3	Kumar	29.0	38000.0	Chennai	Male
4	Divya	28.0	51000.0	Delhi	Female
5	Arun	29.0	70000.0	Mumbai	Male
6	Priya	35.0	55000.0	Delhi	Male
7	Vijay	29.0	51000.0	Chennai	Male

Missing values after cleaning:

```
{'Name': 0, 'Age': 0, 'Salary': 0, 'City': 0, 'Gender': 0}
```

Alternative: dropping every row with a missing value would leave only 1 of 8 rows

Graph

